Metadata for the paper entitled “Flow intermittence alters carbon processing in rivers through chemical diversification of leaf litter” submitted in Limnology and Oceanography Letters

Table 1. Description of the fields needed to describe the creation of your dataset.

Title of dataset	Flow intermittence alters carbon processing in rivers through chemical diversification of leaf litter
URL of dataset	10.6084/m9.figshare.15078972
Abstract	Complete dataset a from the paper "Flow intermittence alters carbon processing in rivers through chemical diversification of leaf litter” published in Limnology & Oceanography Letters. See below for information about the experimental design and methods. More details can be found at the original publication: del Campo R., Corti R, Singer G. 2021. Flow intermittence alters carbon processing in rivers through chemical diversification of leaf litter. Limnology & Oceanography Letters. Manuscript abstract: The dry phase of intermittent rivers promotes the accumulation of leaf litter on various terrestrial and aquatic habitats. This environmental heterogeneity causes a chemical diversification of leaf litter by a range of physical and biological degradation processes acting across the various habitats. After flow resumption, the chemically diversified leaves are mixed and subject to continued decomposition downstream. We hypothesized that the chemical diversification of leaf litter during the dry phase would affect leaf litter decomposition under re-established lotic conditions. Our laboratory treatments mimicking dry-phase habitats caused a strong chemical diversification of leaf litter, which – upon combination in mixed litter bags – accelerated its decomposition in a perennial river reach. We suggest that intermittent rivers may act as hotspots of organic matter diversification, with potential implications for organic matter processing at the river-network scale.
Keywords	Leaf litter decomposition, chemical diversity, intermittent rivers, organic matter dynamics, biodiversity, ecosystem functioning, BEF
Dataset lead author	Rubén del Campo
Position of data author	Post-doctoral researcher
Address of data author during data collection	Leibniz-Institute of Freshwater Ecology and Inland Fisheries (IGB). Department of Ecohydrology. Müggelseedam 310, 12587, Berlin, Germany
Current address of data author	Department of Ecology, University of Innsbruck. Technikerstrasse 25, 6020 Innsbruck (Austria).
Email address of data author	ruben.del-campo@uibk.ac.at; ruben.dlcampo@gmail.com

Primary contact person for dataset	Same as data author.
Position of primary contact person	Same as data author.
Address of primary contact person	Same as data author.
Email address of primary contact person	Same as data author.
Organization associated with the data	Same institutions as detailed at the address of data author.
Usage Rights	Publicly available and free to use.
Geographic region	Laboratory experiments were carried out at IGB-Berlin (Germany). Field experiment was carried out in the Löcknitz river (Brandenburg, Germany).
Geographic coverage	Coordinates of the Löcknitz river: 52°24'43.7"N, 13°49'33.6"E
Temporal coverage - Begin date	01/06/2014
Temporal coverage - End date	01/08/2017
General study design	This study was carried out in two experimental phases. In the first phase, we preconditioned alder leaf litter (*Alnus glutinosa*) under 7 laboratory treatments simulating the accumulation of this leaf litter under various environmental conditions typically found during the dry phase of intermittent rivers. Following preconditioning, sub-samples of all treatments were freeze-dried, ground using a ball mill and analyzed for C- and N-content, other nutrients such as P, Ca, Mg and K, and macromolecular organic C moieties by Fourier-transform infrared spectroscopy (FTIR). In the second phase, we used the preconditioned leaf litter to create fine- and coarse-mesh bags. Litterbags contained leaves of single treatments (7 treatments x 4 replicates) and mixtures of leaves of increasing treatment richness in all possible combinations of 2, 4, and 6 treatments. We incubated all these litterbags in the Löcknitz river for 23 days to measure leaf litter decomposition (mass loss), shredders colonization (shredder density) and fungal biomass (ergosterol). Parallel to the field decomposition experiment, we performed a respiration assay in the laboratory to measure the microbial respiration as a proxy of microbial decomposition activity on leaves from single treatments and mixtures.
Methods description	<u>Leaf litter preconditioning and preparation of mixtures:</u> We collected fresh leaves of *Alnus glutinosa* (alder) directly from several trees along the Löcknitz river and let them air-dry for two weeks. Leaf litter was preconditioned under seven laboratory treatments mimicking the following habitats typical of drying rivers detailed in the Table 1 of the manuscript: Anoxic pools (T1), sunlight (T2), warm, nutrient-rich pools (T3), cold, nutrient-poor pools (T4), moist shaded sediments (T5), alternation between dry (T2) and wet (T3) habitats (T6), and fresh abscission during drought (T7). Afterwards, we prepared fine- and coarse-mesh bags (0.5 and 8 mm, respectively, 15 x 15 cm size) containing leaves of single treatments (7 treatments x 4 replicates) and mixtures of leaves of increasing treatment richness in all possible combinations of 2, 4, and 6 treatments. We filled each litterbag with 12 leaves. In mixtures, the 12 leaves were evenly partitioned across the component treatments. All leaves were scanned prior to bag assembly to later measure treatment-specific leaf areas by

	digital image analysis (ImageJ, https://imagej.nih.gov/ij/) and compute the exact contributions of component treatments on a dry mass (DM) basis. For this, we established conversion factors of leaf area to DM (48h, 105 °C) for each treatment from 20 leaves.
	Aquatic decomposition experiment: To measure aquatic decomposition of single treatments and mixtures, we incubated all litterbags in the Löcknitz River for 23 days in August 2014. Litterbags were tied to iron rods and fixed on the riverbed in four reaches of 50 m with running water and homogeneous substrate, depth and flow conditions.
	Microbial respiration assay: We incubated 12 leaf discs by mixture or single treatment in 250 mL sealed bottles filled with mineral water (Volvic) at room temperature in a water bath. As microbial inoculum we used 10 mL of river water filtered by 0.7 µm pre-combusted glass fiber filters (Whatman GF/F, Maidstone, UK). Dissolved oxygen concentrations were measured 13 times over 24 days with a needle-based micro-optode (PM-PSt7 on a Microx 4 trace meter; PreSens, Germany). 10 bottles were filled with plain water as a control. Oxygen consumption rates (day-1) were computed as first order oxygen decay rates from log-linear regression models.
Laboratory, field, or other analytical methods	Leaf litter mass loss: The leaves from each litterbag were dried (105 °C, 48h), weighed, and leaf litter mass loss computed as the difference between initial and final litterbag DM divided by initial DM.
	Leaf litter chemical composition: Following preconditioning, sub-samples of all treatments were freeze-dried, ground using a ball mill and analyzed for - C and N elemental composition: Elementar vario EL C/N elemental analyzer (Germany) - Content in P, Ca, Mg and K: ICP-OES (Thermo Scientific, iCAP 6500, USA). - Macromolecular organic C moieties by Fourier-transform infrared spectroscopy (FTIR): 1.75 mg (± 0.05 mg) of ground, freeze-dried leaf litter were mixed with 400 mg of KBr, homogenized in a ball mill for 30 s and pressed to a pellet. FTIR spectra (as absorbance in units of cm^{-1}) were measured for wavenumbers of 400 to 4000 cm^{-1} at a resolution of 4 cm^{-1} with 200 scans per sample on a FTIR-8300 (Shimadzu, Japan). We subtracted a blank spectrum (pure KBr pellet) measured as a background every 5 samples. We normalized the raw spectra by dividing each peak by the square root of the sum of all peaks, and corrected the baseline drift by the iterative restricted least squares method. Then, we extracted heights of seventeen peaks from 800 to 1800 cm^{-1} corresponding to main functional groups of carbohydrates, cellulose, lignin and phenolic compounds.
	Shredder density: Coarse-mesh bags were washed individually in the laboratory with tap water above a 250 µm sieve to collect invertebrates, which were preserved in 70% ethanol. Individuals were counted, identified to family level and classified by guilds. The density of shredders was expressed as number of individuals per DM of leaf litter.

	Fungal biomass: From leaves in fine-mesh bags we cut a set of 12 discs with a cork borer (10 mm) to measure fungal biomass as ergosterol. Leaf discs were frozen at -80 °C pending lyophilization and weighing. Lipids were extracted using a KOH-methanol solution at 80 °C for 30 min and purified using solid-phase extraction cartridges (Waters Sep-Pak®, Vac RC, 500 mg, C18 cartridges, Waters Corp., USA). Ergosterol was eluted using isopropanol and quantified by HPLC with absorbance detection at 282 nm (Dionex UltiMate 3000 LC, USA). Values of ergosterol were expressed as µg g^{-1} DM.
	Microbial respiration: Explained above in "Method description".
Quality control	All data and code were double-checked by 2 people.
Additional information	Estimation of chemical composition and chemical diversity in leaf litter mixtures: We combined information from FTIR peaks and elemental analysis of the preconditioning treatments to perform a principal component analyses (PCA) using z-standardized data. We estimated the chemical diversity of leaf litter mixtures as Rao's quadratic entropy. For each mixture, RaoQ was computed as the mean pairwise Euclidean distance between component treatment centroids based on their scores on the first two PCA-axes, weighted by the relative abundances of each treatment. RaoQ equals 0 for litterbags containing single treatments. To estimate the chemical composition of leaf mixtures we used community-weighted means. This way we obtained average values of each chemical trait (FTIR peaks, nutrients) from their component treatment, weighted by the relative abundances of each treatment in the mixture. Weighted means of mixtures were combined with single treatment data to run a second PCA. Average scores of each single treatment or mixture on the first two PCA axes served as a 2-dimensional proxy of chemical composition. Estimation of expected values for response variables: We estimated expected values of all response variables for each leaf litter mixture using observed values in single treatments, and compared those to observed values in mixtures. Expected values were computed as the weighted average of observed values of component treatments on a DM basis.

Table 2. Description of the variables (i.e., columns) in the dataset called "raw_leaf_litter_chemical_composition.txt". This matrix describes the chemical composition of leaf litter exposed to the 7 preconditioning treatments.

Column name	Definition	Units
habitat	Preconditioning habitat description	Factor
treatment	Preconditioning treatment identity (1 to 7)	Factor
N	Nitrogen	g/100g
C	Carbon	g/100g
Ca	Calcium	g/100g
K	Potassium	g/100g
Mg	Magnesium	g/100g

P	Phosphorous	g/100g
CBH	FTIR peak at 901 nm considered as signal of carbohydrates	Relative intensity
CBH2	FTIR peak at 874 nm considered as signal of carbohydrates	Relative intensity
CBH3	FTIR peak at 851 nm considered as signal of carbohydrates	Relative intensity
Cel	FTIR peak at 1040 nm considered as signal of cellulosic compounds	Relative intensity
Cel2	FTIR peak at 1130 nm considered as signal of cellulosic compounds	Relative intensity
Cel3	FTIR peak at 1161 nm considered as signal of cellulosic compounds	Relative intensity
Lig	FTIR peak at 1229 nm considered as signal of lignin-like compounds	Relative intensity
Lig2	FTIR peak at 1265 nm considered as signal of lignin-like compounds	Relative intensity
Lig3	FTIR peak at 1315 nm considered as signal of lignin-like compounds	Relative intensity
Lig4	FTIR peak at 1375 nm considered as signal of lignin-like compounds	Relative intensity
Lig5	FTIR peak at 1450 nm considered as signal of lignin-like compounds	Relative intensity
Lig6	FTIR peak at 1514 nm considered as signal of lignin-like compounds	Relative intensity
Lig7	FTIR peak at 1551 nm considered as signal of lignin-like compounds	Relative intensity
Prot	FTIR peak at 1595 nm considered as signal of protein-like compounds	Relative intensity
Tann	FTIR peak at 1620 nm considered as signal of phenolic compounds	Relative intensity
Phen-Prot	FTIR peak at 1653 nm considered as signal of phenolic compounds	Relative intensity
CN	Ratio C:N	Dimensionless
CP	Ratio C:P	Dimensionless
NP	Ratio N:P	Dimensionless

Table 3. Description of the variables (i.e., columns) in the dataset called: “raw_leaf_litter_decomposition.txt”. Matrix contains information about leaf litter mass loss, fungal biomass (ergosterol) and shredder density.

Column name	**Definition**	**Units**
label	Label code of each litterbag	string
llb_type	Type of litterbag (fine- or coarse-mesh bag)	string
treat_richness	Number of treatment combinations (1,2,4, or 6) in the litterbag. E.g. 1 = litterbag with a single treatment, 6 = litterbag with a mixture of 6 different treatments	factor
single_treat	Identification of treatments in litterbags with single treatments. NA for mixtures.	factor
treat_combi	Identification of the treatments combined in the litter bag E.g. 111222333444 = litterbag containing 3 leaves of treatments T1, T2, T3 and T4. Each litterbag contains 12 leaves in total.	string

mass_loss	Leaf litter mass loss observed in fine- and coarse-mesh bags during the aquatic decomposition experiment.	% of dry mass
ergosterol	Ergosterol content measured in leaf litter from fine-mesh bags as a proxy of fungal biomass.	µg of ergosterol per gram of leaf litter dry mass
shredder_density	Density of shredders found in coarse-mesh bags.	Individuals per gram of leaf litter dry mass

Table 4. "raw_macroinvertebrate_community.txt". Species by site matrix containing abundance data of all macroinvertebrates identified in coarse-mesh bags.

Column name	Definition	Units
Taxa	Each row corresponds to a macroinvertebrate taxon.	String
Samples	Each column corresponds to a litterbag sample. Number code corresponds to variable "Label" in the rest of matrixes	Abundance data is expressed as number of individuals.

Table 5. Description of the variables (i.e., columns) in the dataset called: "raw_microbial_respiration.txt". This matrix contains microbial respiration rates computed from the laboratory bottle respiration assay.

Column name	Definition	Units
Number	Number code of the bottle used during the microbial respiration assay for each sample	string
treat_richness	Number of treatment combinations (1,2,4, or 6) in the litterbag. E.g. 1 = litterbag with a single treatment, 6 = litterbag with a mixture of 6 different treatments	factor
single_treat	Identification of treatments in litterbags with single treatments. NA for mixtures.	factor
treat_combi	Identification of the treatments combined in the litter bag E.g. 111222333444 = litterbag containing 3 leaves of treatments T1, T2, T3 and T4. Each litterbag contains 12 leaves in total.	string
respiration	Microbial respiration rate computed as first order oxygen decay rates from log-linear regression models.	Day-1

Table 6. Description of the variables (i.e., columns) in the dataset called "raw_mixture_fractions.txt". Dry mass (48h, 105 °C) of leaf litter from each treatment composing each litterbag. Used to correct RaoQ & CWM by fractions biomass. Matrix used to correct RaoQ and Community-weighed means by the dry biomass of each fractions composing each leaf mixture.

Column name	Definition	Units
label	Label code of each litterbag	string
T1	Dry mass of leaf litter from treatment T1	g
T2	Dry mass of leaf litter from treatment T2	g
T3	Dry mass of leaf litter from treatment T3	g
T4	Dry mass of leaf litter from treatment T4	g
T5	Dry mass of leaf litter from treatment T5	g
T6	Dry mass of leaf litter from treatment T6	g

T7	Dry mass of leaf litter from treatment T7	g
llb_type	Type of litterbag (fine- or coarse-mesh bag)	string
treat_richness	Number of treatment combinations (1,2,4, or 6) in the litterbag. E.g. 1 = litterbag with a single treatment, 6 = litterbag with a mixture of 6 different treatments	factor
single_treat	Identification of treatments in litterbags with single treatments. NA for mixtures.	factor
treat_combi	Identification of the treatments combined in the litter bag E.g. 111222333444 = litterbag containing 3 leaves of treatments T1, T2, T3 and T4. Each litterbag contains 12 leaves in total.	string

Table 7. Description of the variables (i.e., columns) in the dataset called "leaf_litter_chemical_composition_&_diversity.txt". Information about chemical composition and chemical diversity of litterbags (single treatments and leaf mixtures).

Column name	Definition	Units
label	Label code of each litterbag	string
llb_type	Type of litterbag (fine- or coarse-mesh bag)	string
treat_richness	Number of treatment combinations (1,2,4, or 6) in the litterbag. E.g. 1 = litterbag with a single treatment, 6 = litterbag with a mixture of 6 different treatments	factor
single_treat	Identification of treatments in litterbags with single treatments. NA for mixtures.	factor
treat_combi	Identification of the treatments combined in the litter bag E.g. 111222333444 = litterbag containing 3 leaves of treatments T1, T2, T3 and T4. Each litterbag contains 12 leaves in total.	string
RAOQ	Rao's quadratic entropy computed as a proxy of the chemical diversity of leaf litter. See addition information for a more detailed description.	Dimensionless
PC1	PC1 scores based on community-weighted means of chemical traits for each litterbag. See addition information for a more detailed description.	Dimensionless
PC2	PC2 scores based on community-weighted means of chemical traits for each litterbag. See addition information for a more detailed description.	Dimensionless

Table 8. Description of the variables (i.e., columns) in the dataset called: "leaf_litter_decomposition_gamlss.txt". This matrix contains observed and expected results of all decomposition response variables (leaf mass loss, fungal biomass and shredder density). This data is used to run GAMLSS models to analyse the decomposition response to the treatment richness gradient of leaf mixtures.

Column name	Definition	Units
label	Label code of each litterbag	string
llb_type	T8ype of litterbag (fine- or coarse-mesh bag)	string
treat_richness	Number of treatment combinations (1,2,4, or 6) in the litterbag. E.g. 1 = litterbag with a single treatment, 6 = litterbag with a mixture of 6 different treatments	factor

single_treat	Identification of treatments in litterbags with single treatments. NA for mixtures.	factor
treat_combi	Identification of the treatments combined in the litter bag E.g. 111222333444 = litterbag containing 3 leaves of treatments T1, T2, T3 and T4. Each litterbag contains 12 leaves in total.	string
mass_loss	Leaf litter mass loss observed in fine- and coarse-mesh bags during the aquatic decomposition experiment.	% of dry mass
ergosterol	Ergosterol content measured in leaf litter from fine-mesh bags as a proxy of fungal biomass.	µg of ergosterol per gram of leaf litter dry mass
shredder_density	Density of shredders found in coarse-mesh bags.	Individuals per gram of leaf litter dry mass
exp_mass_loss	Expected value of mass loss computed from observed values (see additional information).	% of dry mass
exp_ergosterol	Expected value of ergosterol computed from observed values (see additional information).	µg of ergosterol for gram of leaf litter dry mass
exp_shredder_density	Expected value of shredder density computed from observed values (see additional information).	Individuals per gram of leaf litter dry mass

Table 9. Description of the variables (i.e., columns) in the dataset called: "microbial_respiration_gamlss.txt". This matrix contains observed and expected results for the microbial respiration rate This data is used to run GAMLSS models to analyse the response of microbial respiration to the treatment richness gradient of leaf mixtures.

Column name	**Definition**	**Units**
Number	Number code of the bottle used during the microbial respiration assay for each sample	string
treat_richness	Number of treatment combinations (1,2,4, or 6) in the litterbag. E.g. 1 = litterbag with a single treatment, 6 = litterbag with a mixture of 6 different treatments	factor
single_treat	Identification of treatments in litterbags with single treatments. NA for mixtures.	factor
treat_combi	Identification of the treatments combined in the litter bag E.g. 111222333444 = litterbag containing 3 leaves of treatments T1, T2, T3 and T4. Each litterbag contains 12 leaves in total.	string
respiration	Microbial respiration rate computed as first order oxygen decay rates from log-linear regression models.	Day-1
exp_respiration	Expected value of microbial respiration computed from observed values (see additional information).	Day-1